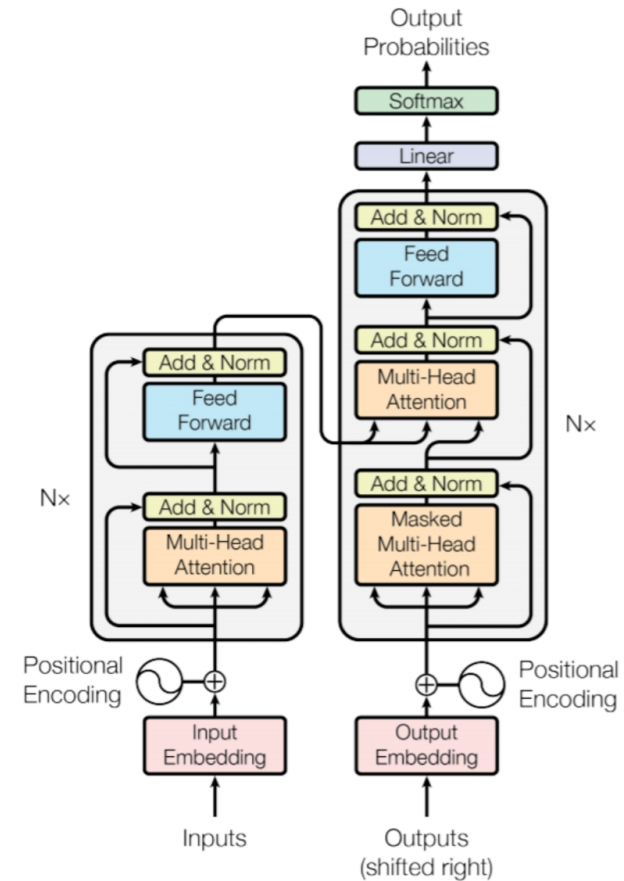


Computer Vision and Deep Learning

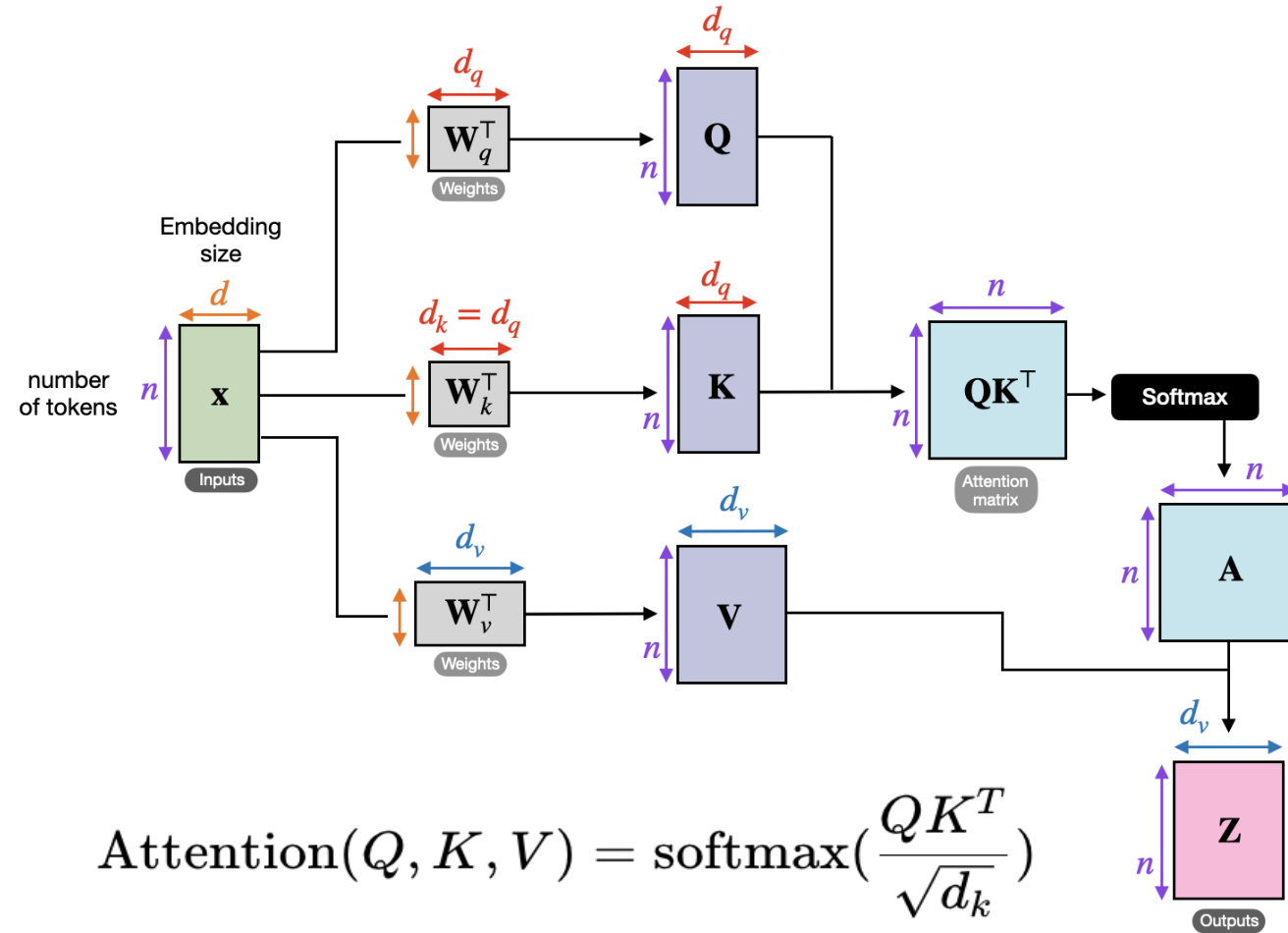
Lab 06: Vision Transformers (ViT)

Transformers

- Introduced in the "Attention is All You need Paper" to solve NLP tasks.
- Describe the effective application of self-attention operator to address non local relationship for language modeling.
- The model architecture is designed in two macro structures: the Encoder and Decoder.

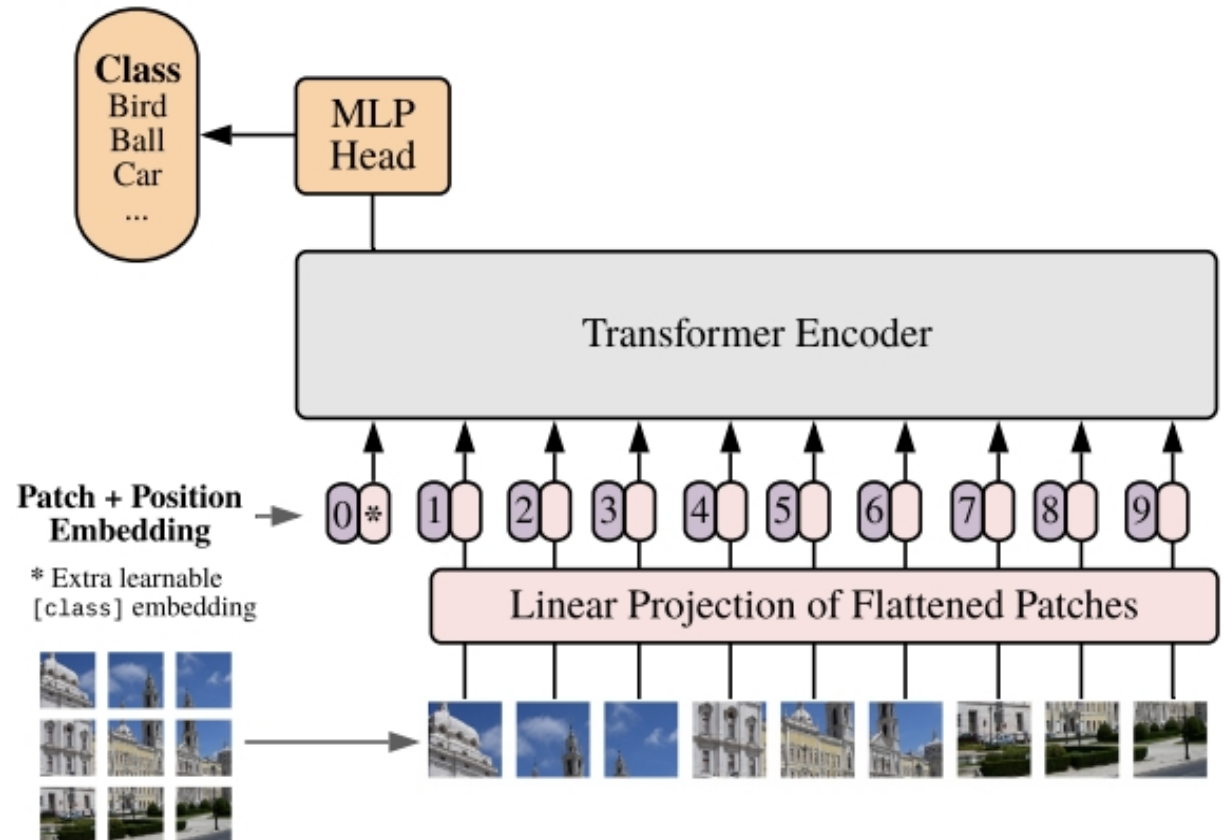


Self-Attention



Vision Transformer (ViT)

- Uses only the encoder part of the transformers in order to perform features extraction
- 2D positional embedding for spatial information reference.
- Tokens are extracting by a liner layer over non-overlapping windows. Not using pixel tokens because of **complexity**
- Additional token embedding [CLS] responsible to learn the global information in the image, i.e. the class.



ViT: 2D Images to Sequence Patches



- Convert the 2D Image to a sequence of patches

ViT: 2D Images to Sequence Patches



Image size: 96x96

Patch size: 32x32

Number of tokens: $3 \times 3 = 9$

Dimension of each token: $3 \times 32 \times 32 = 3072$

- Convert the 2D Image to a sequence of patches

ViT: 2D Images to Sequence Patches

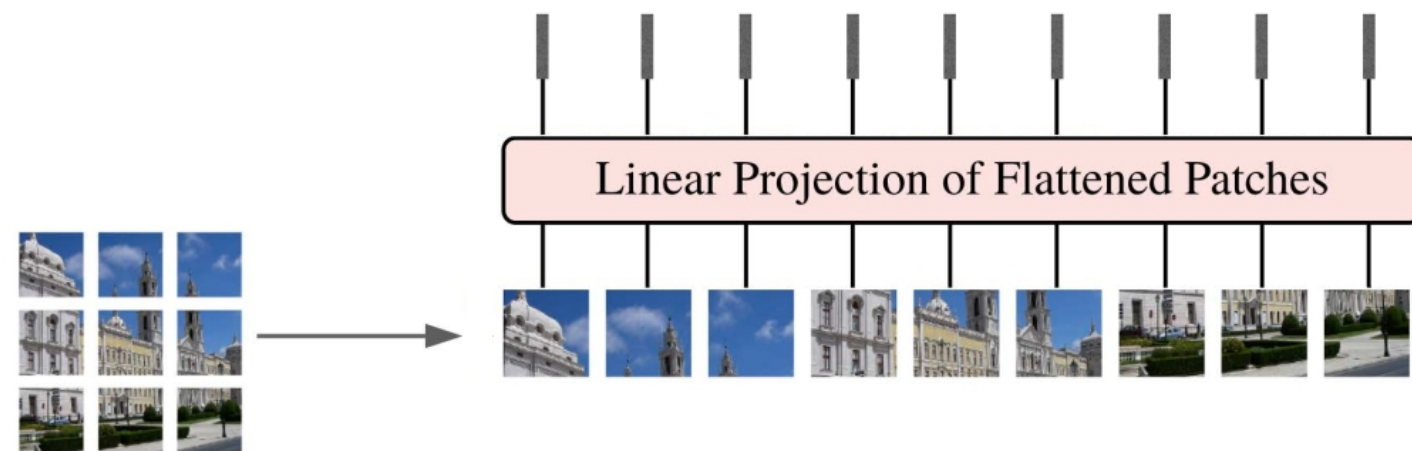
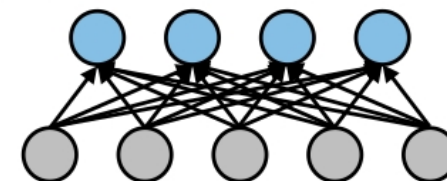


Image size: 96x96
Patch size: 32x32

Number of tokens: $3 \times 3 = 9$
Dimension of each token: $3 \times 32 \times 32 = 3072$

Output dim = hidden size of ViT = 768

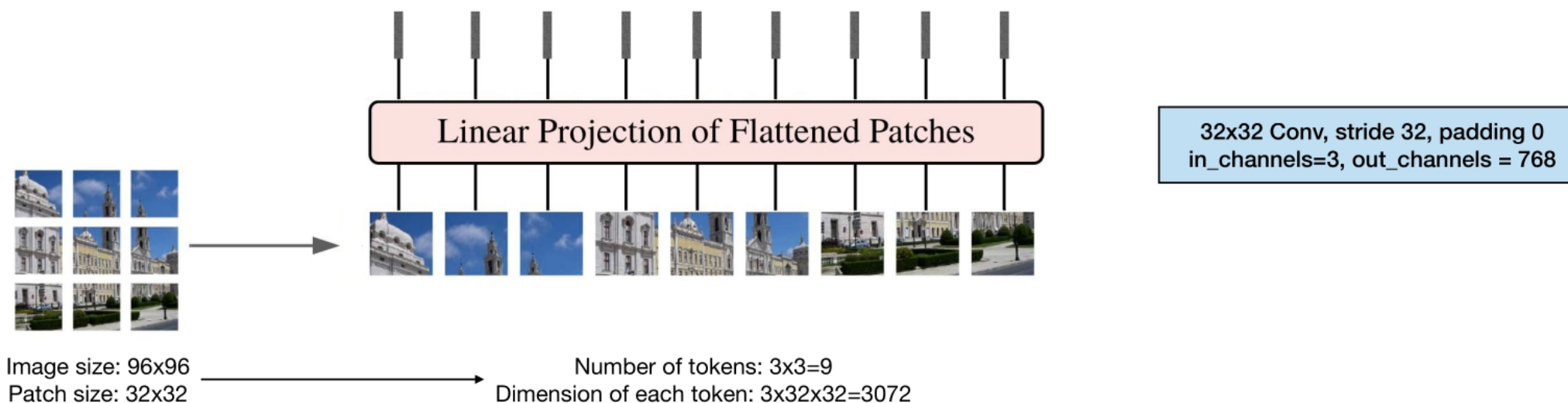


Input dim = 3072

#Parameters: $3072 \times 768 \rightarrow 2.36\text{M}$

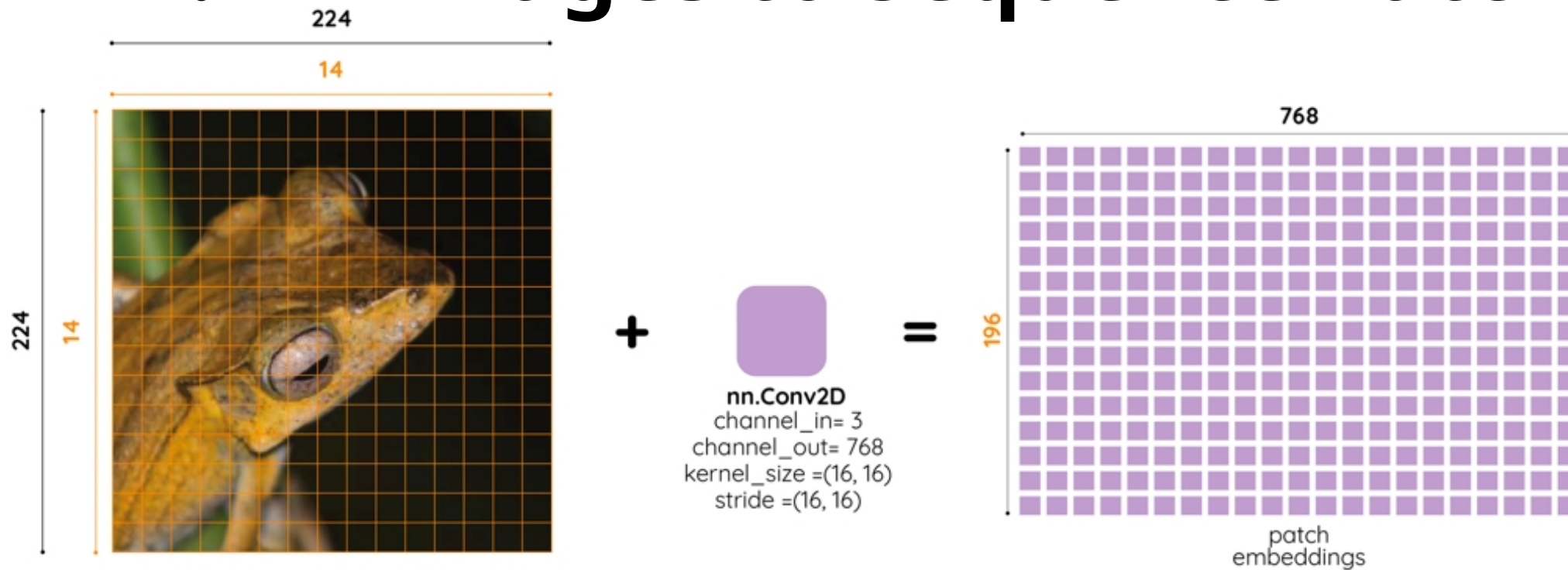
- Convert the 2D Image to a sequence of patches

ViT: 2D Images to Sequence Patches



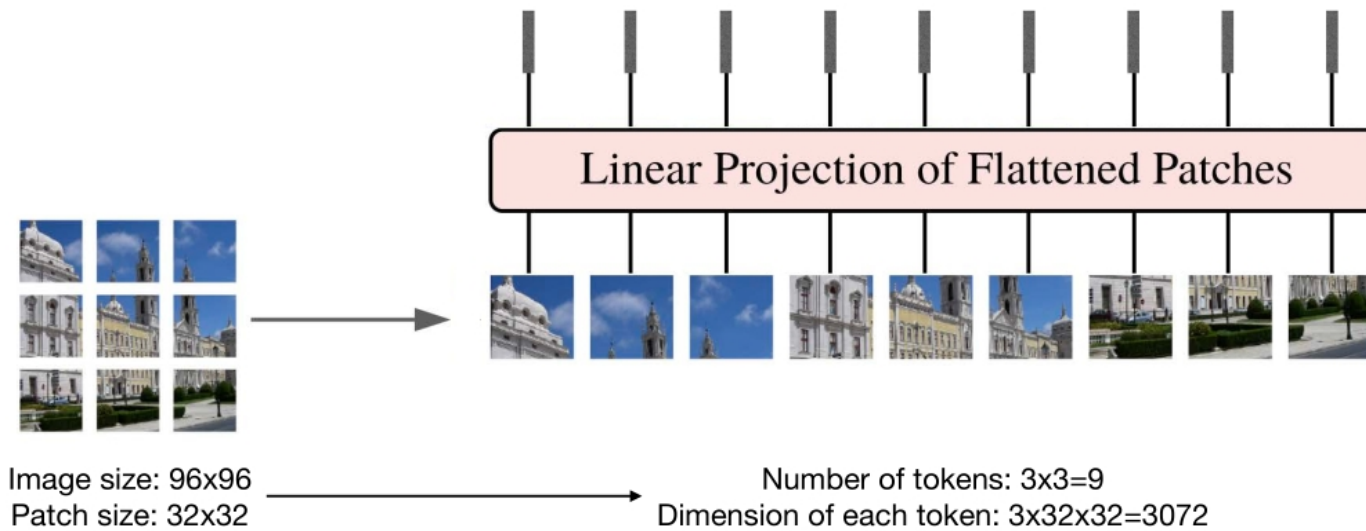
- Convert the 2D Image to a sequence of patches

ViT: 2D Images to Sequence Patches

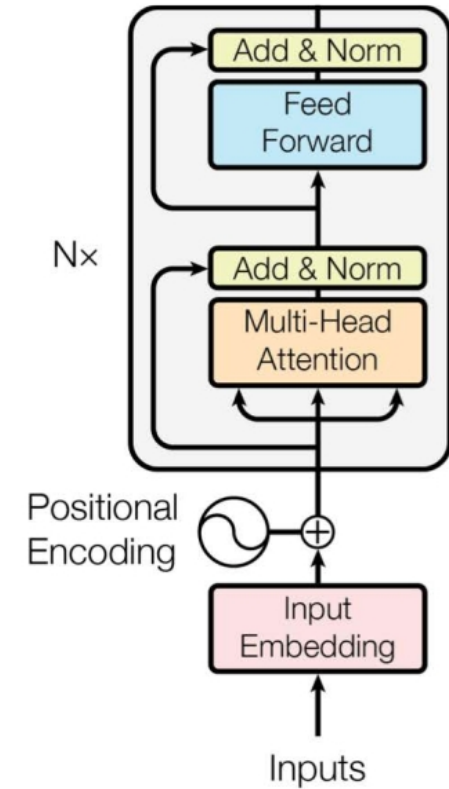


```
x = torch.randn(1, 3, 224, 224)
# 2D conv
conv = nn.Conv2d(3, 768, 16, 16)
conv(x).reshape(-1, 196).transpose(0,1).shape
```

ViT



- Convert the 2D Image to a sequence of patches
- Feed patch embeddings to the standard transformer encoder



ViT: Model Variants

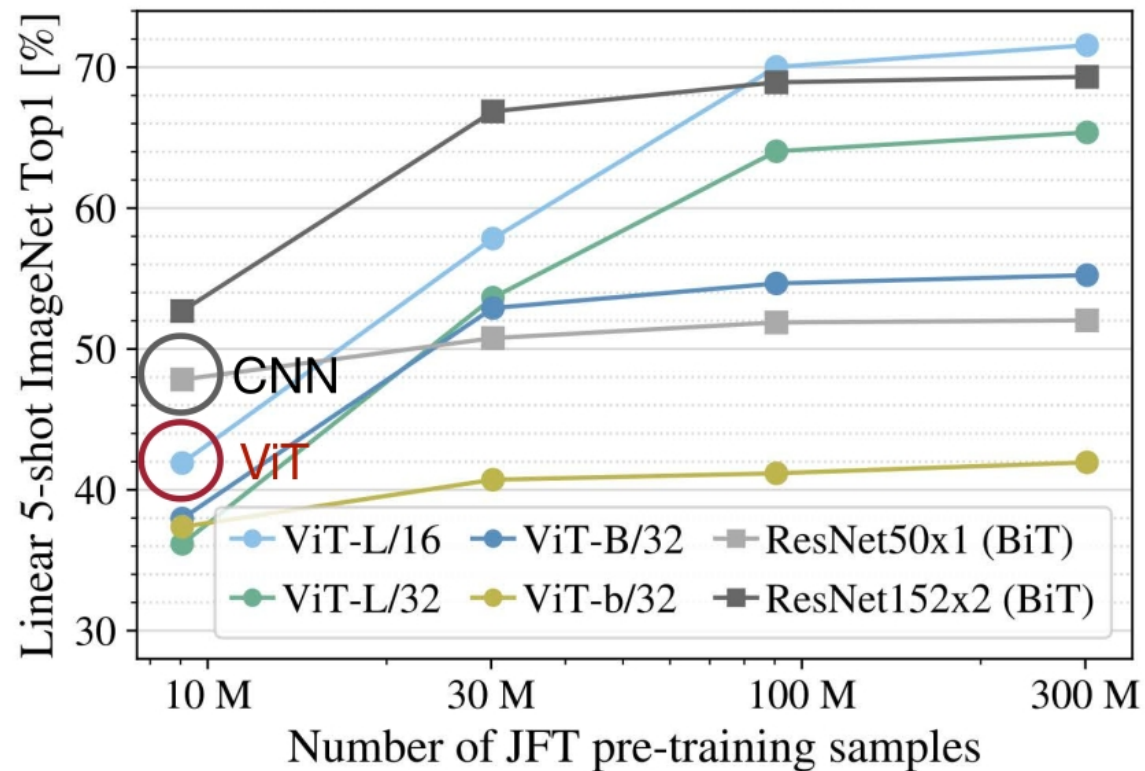
Model	Layers	Hidden size D	MLP size	Heads	Params
ViT-Base	12	768	3072	12	86M
ViT-Large	24	1024	4096	16	307M
ViT-Huge	32	1280	5120	16	632M



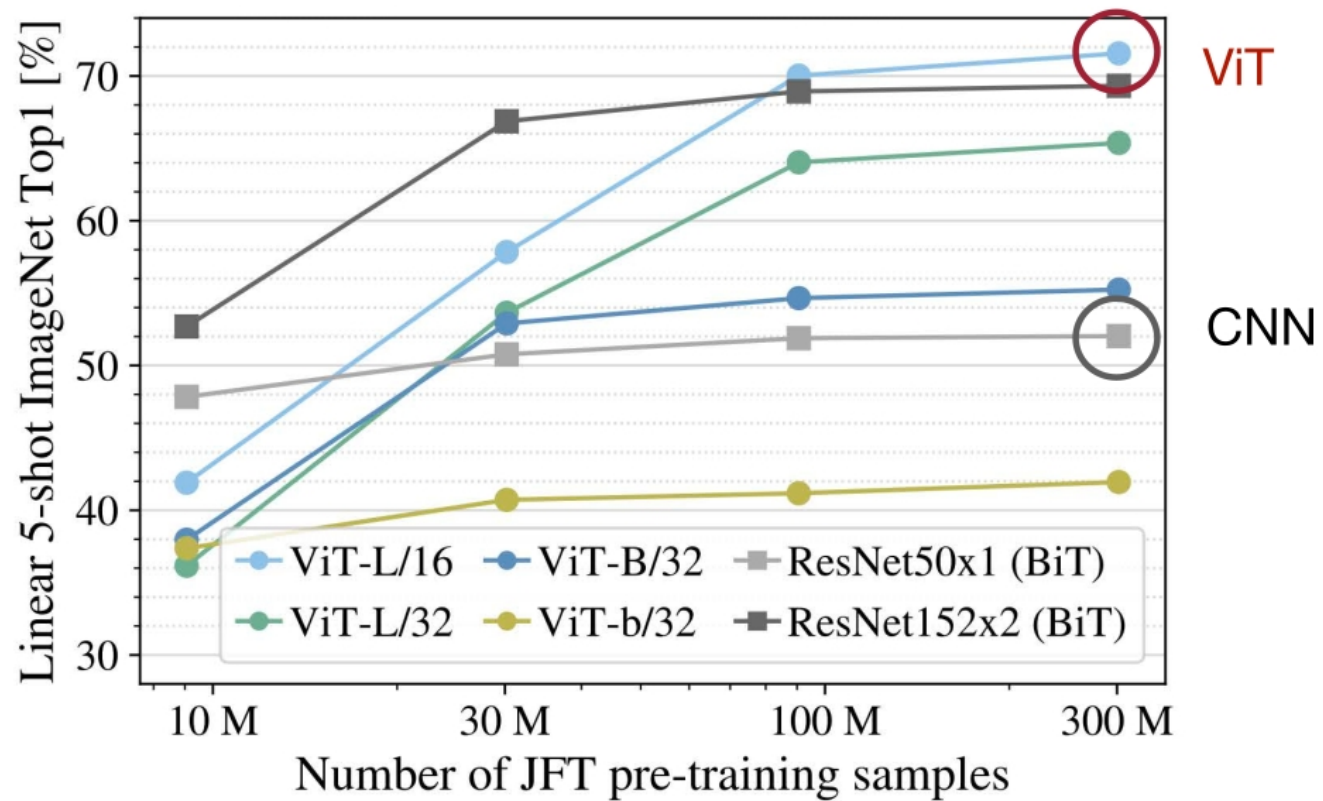
Patch Size:
2, 4, 8, 16, 32, ...

ViT-L/16: ViT-Large with patch size 16x16
resulting in 196 (i.e. 14x14) patches

ViT: Performance



ViT: Performance



Exercises